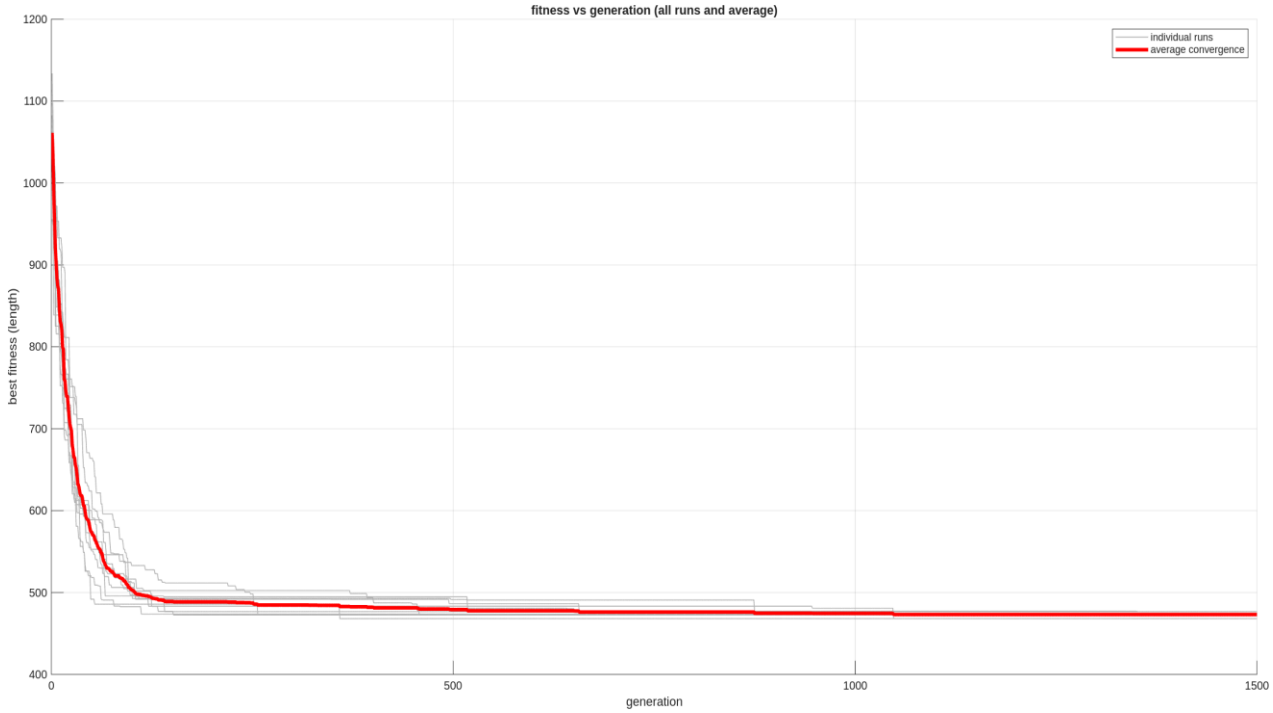
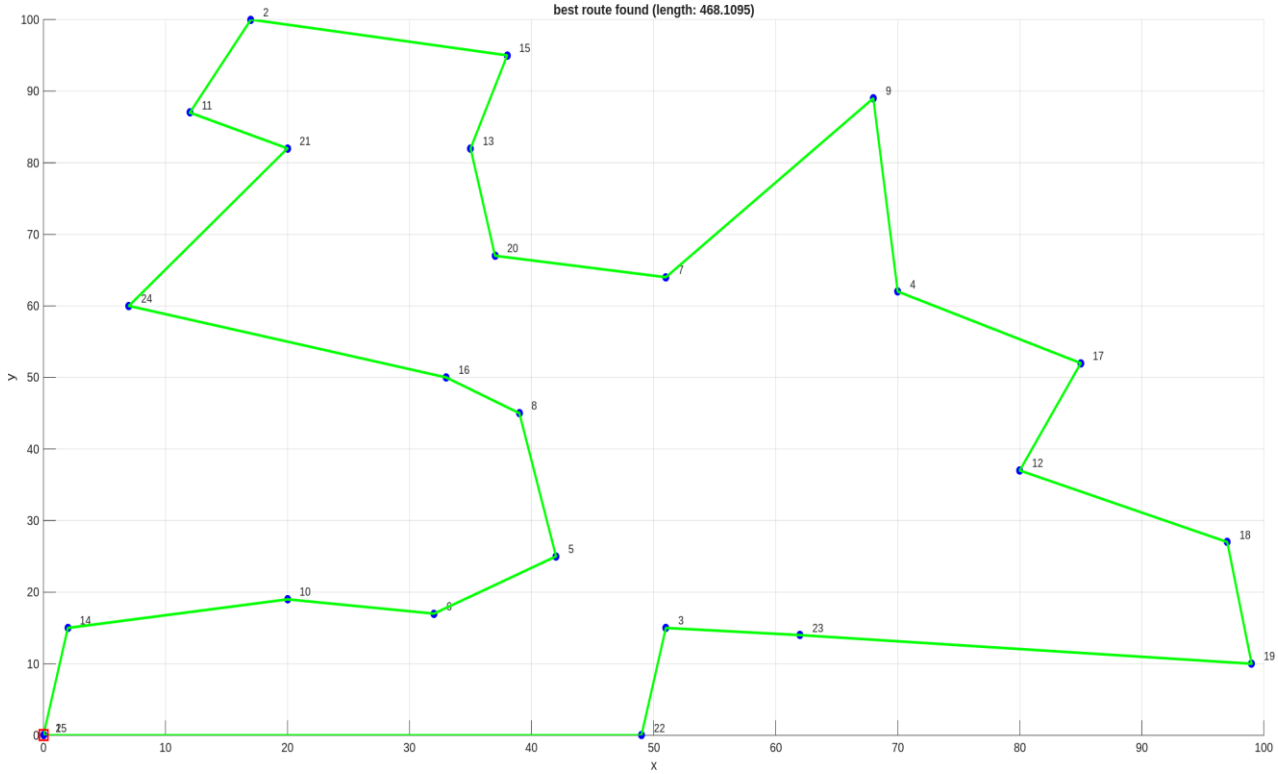
Umint cv3

Best

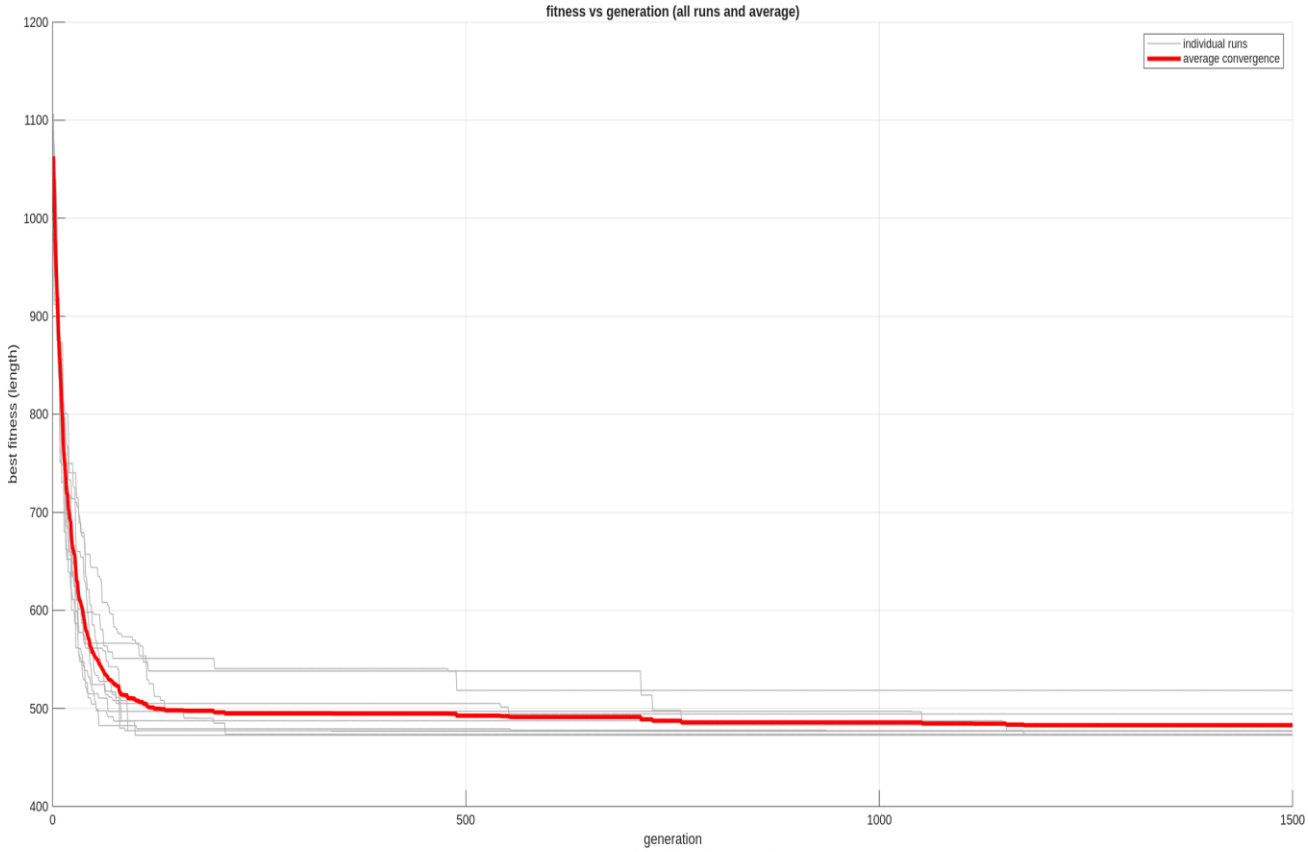


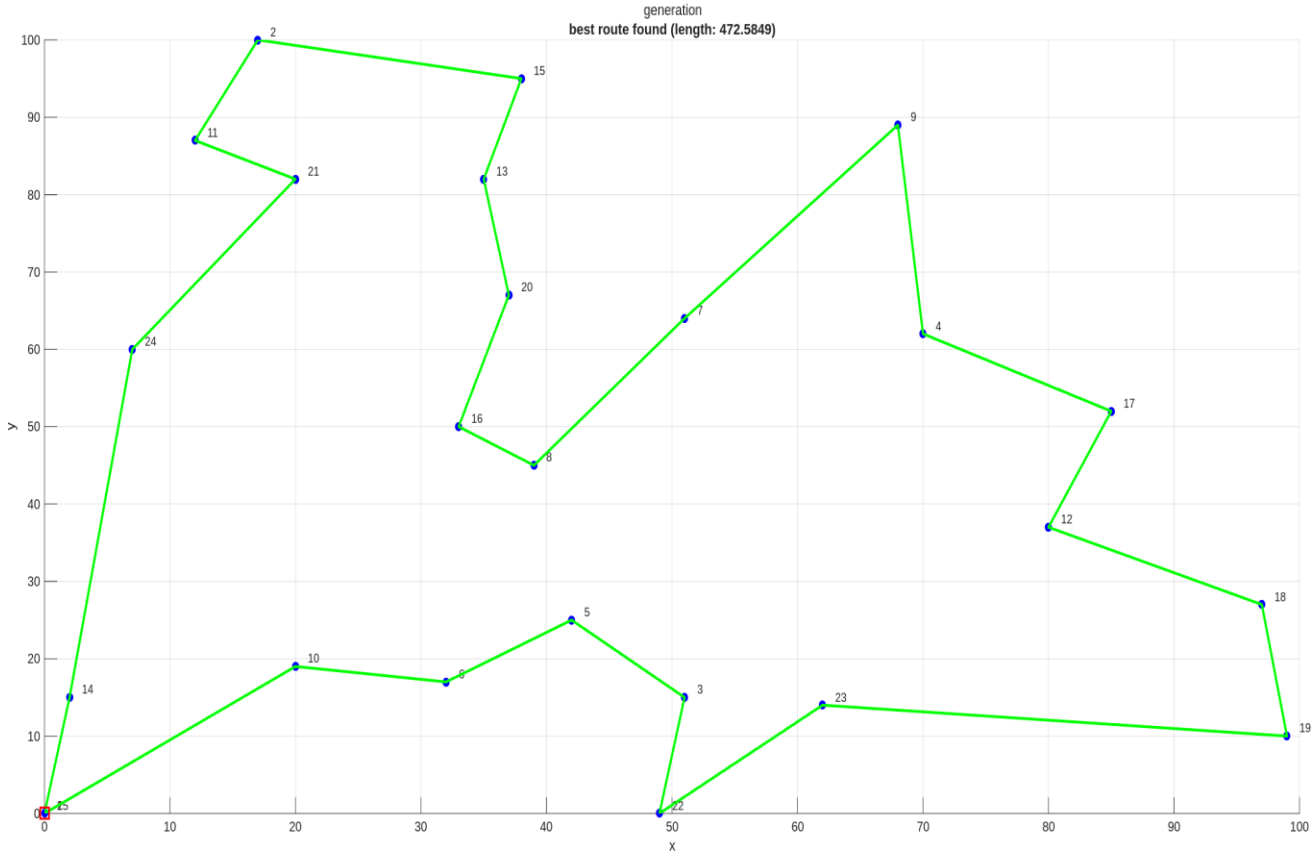



```
popSize = 100;
```

```
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 1500;
numRuns = 10;


starting calculations (10 runs)...
run 1 finished. best fitness: 472.5849
run 2 finished. best fitness: 476.6712
run 3 finished. best fitness: 476.6712
run 4 finished. best fitness: 472.5849
run 5 finished. best fitness: 473.7614
run 6 finished. best fitness: 472.5849
run 7 finished. best fitness: 468.1095
run 8 finished. best fitness: 473.7614
run 9 finished. best fitness: 476.6712
run 10 finished. best fitness: 468.1095

==================== results ====================
absolute best route length: 468.109452
number of successful runs (<= 480): 10 of 10

best route sequence (point indices):
1 22 3 23 19 18 12 17 4 9 7 20 13 15 2 11 21 24 16 8 5 6 10 14 25
```

Without invord

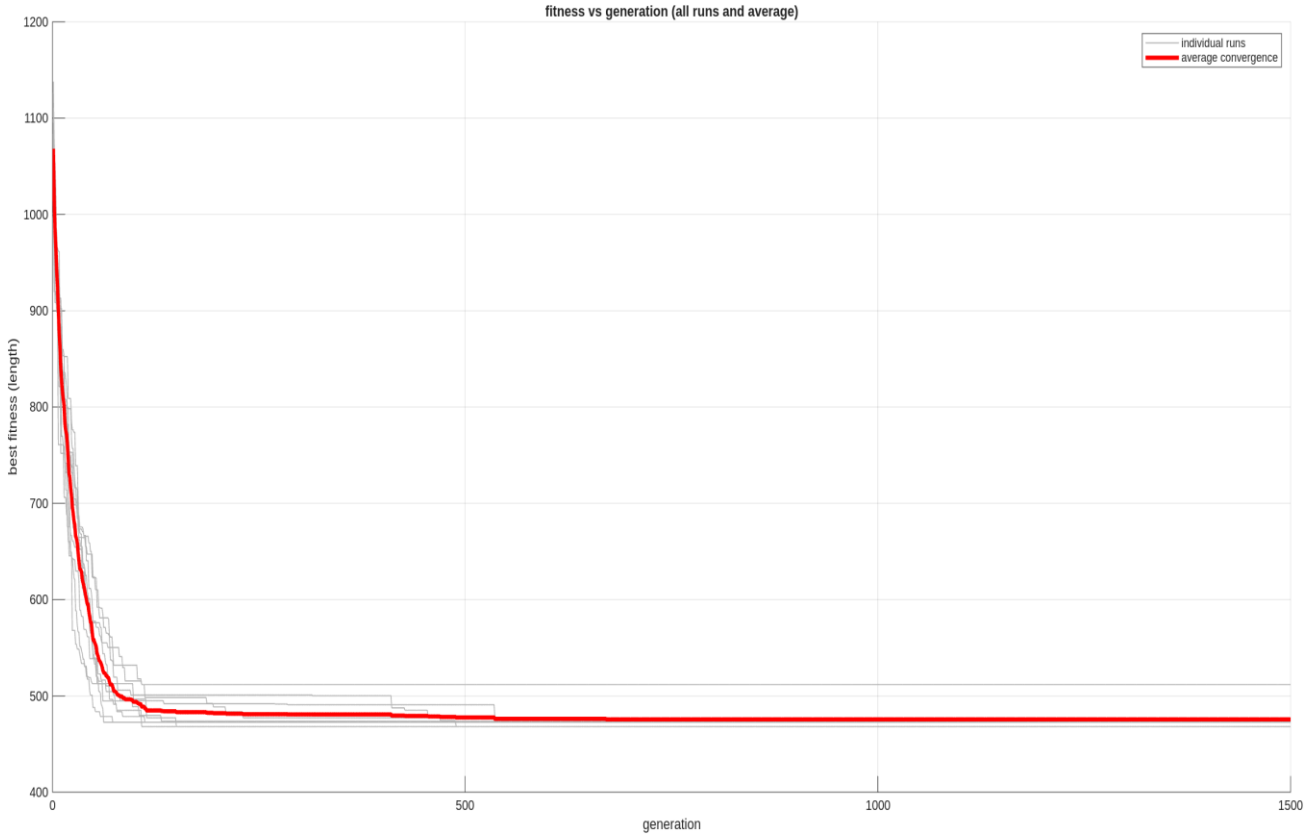

fitness vs generation (all runs and average)
individual runs
average convergence
best fitness (length)
generation
1200
1100
1000
900
800
700
600
500
400
0
500
1000
1500

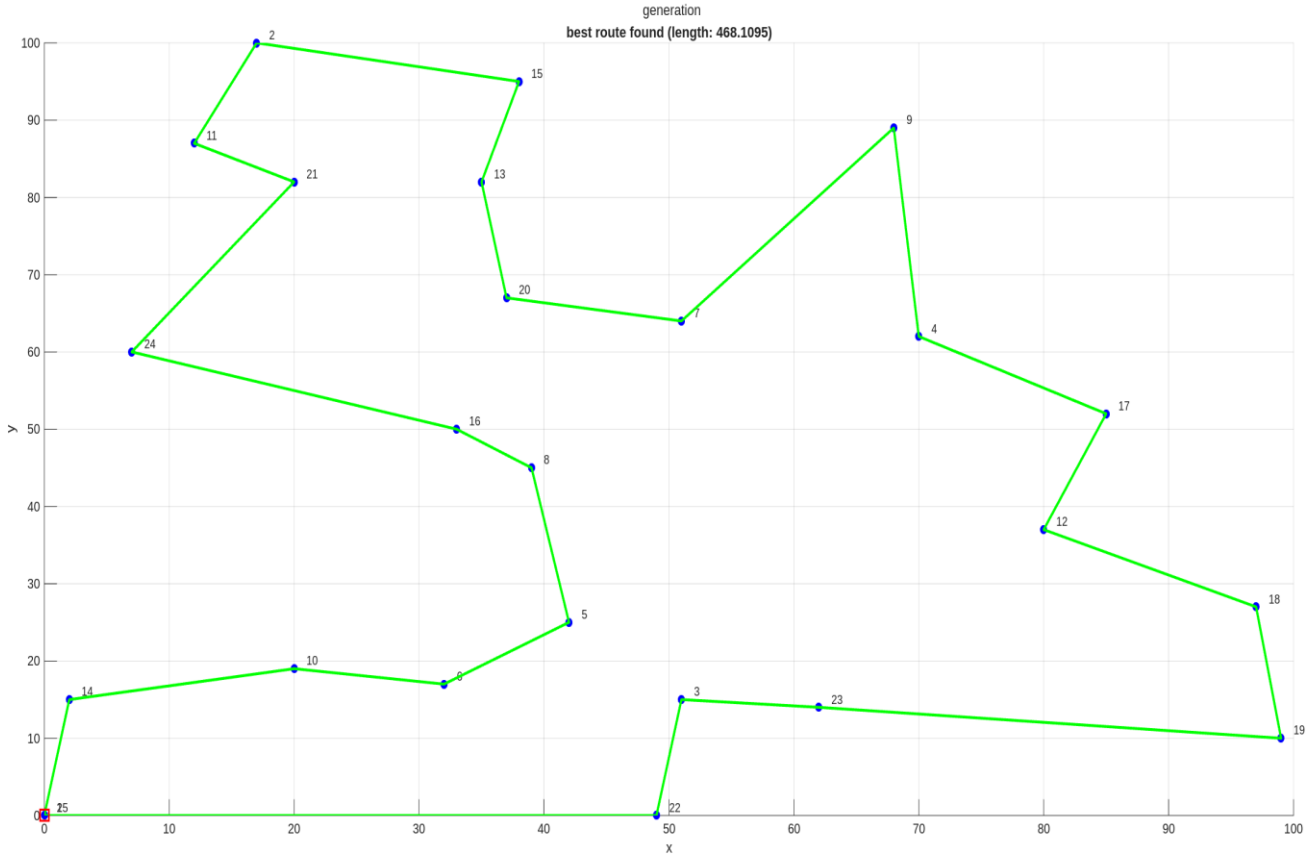

best route found (length: 472.5849)
x
y
2
15
11
21
13
9
20
7
4
24
17
16
8
12
18
5
10
6
3
23
19
14
25
22

```
popSize = 100;
```

```
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 1500;
numRuns = 10;

starting calculations (10 runs)...
run 1 finished. best fitness: 476.6712
run 2 finished. best fitness: 472.5849
run 3 finished. best fitness: 472.5849
run 4 finished. best fitness: 483.1013
run 5 finished. best fitness: 494.2652
run 6 finished. best fitness: 518.4214
run 7 finished. best fitness: 476.6712
run 8 finished. best fitness: 477.1801
run 9 finished. best fitness: 473.7614
run 10 finished. best fitness: 483.3271

==================== results ====================
absolute best route length: 472.584862
number of successful runs (<= 480): 6 of 10
```

Without swappart

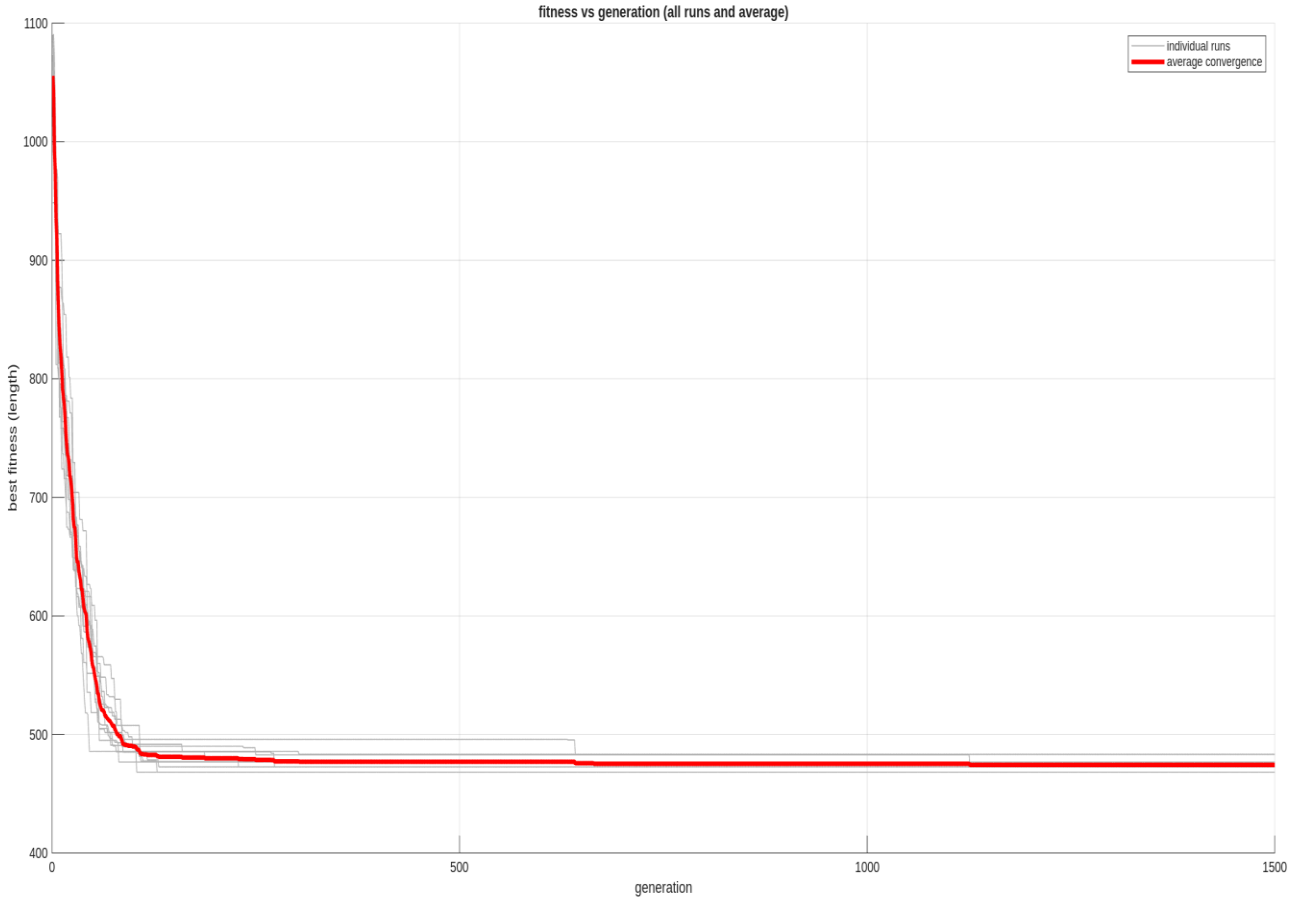


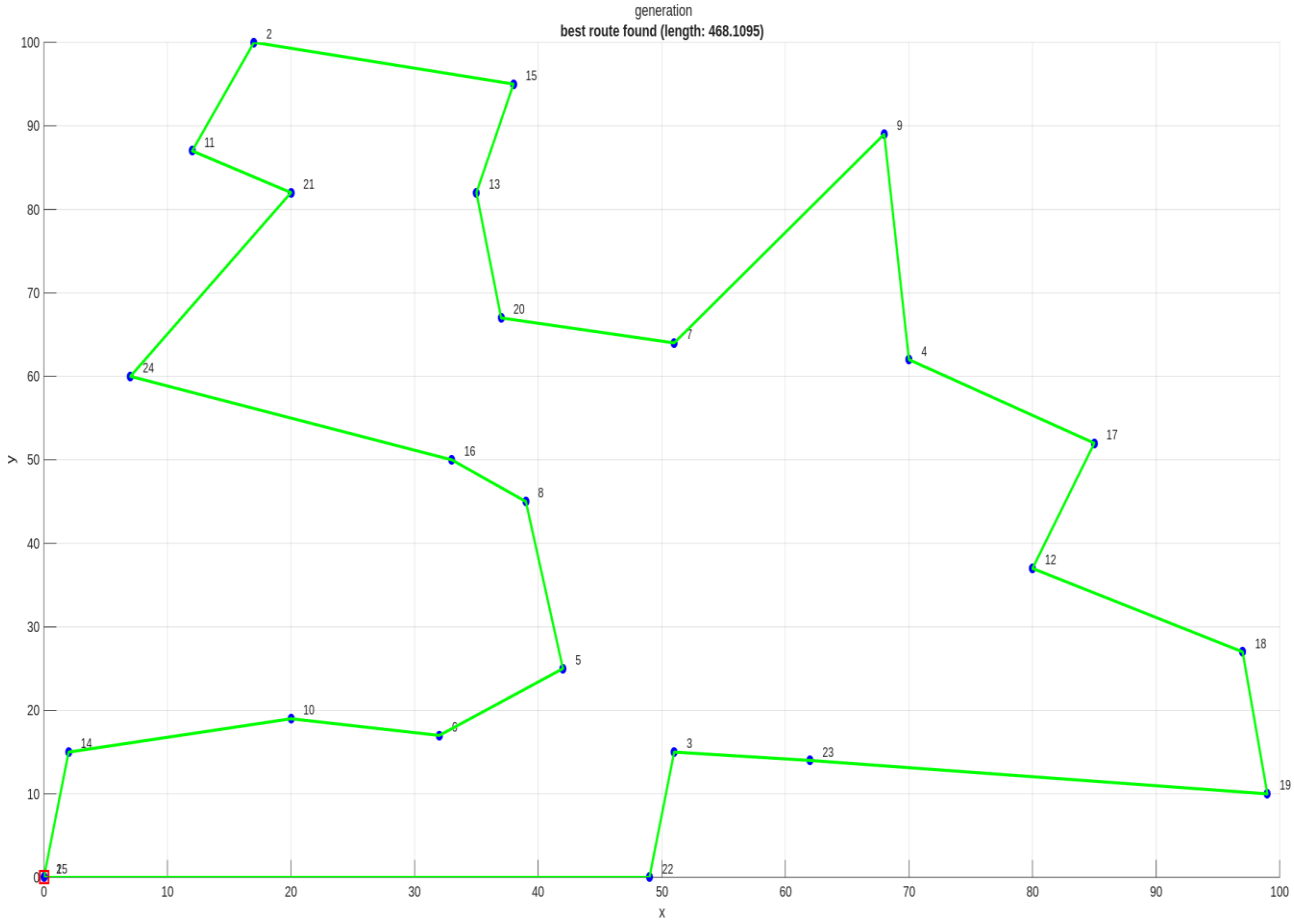



```
popSize = 100;
elitismCount = 5;
mutationRate = 0.12;
```

```
numGenerations = 1500;
numRuns = 10;

starting calculations (10 runs)...
run 1 finished. best fitness: 472.5849
run 2 finished. best fitness: 468.1095
run 3 finished. best fitness: 476.6712
run 4 finished. best fitness: 511.6801
run 5 finished. best fitness: 468.1095
run 6 finished. best fitness: 472.5849
run 7 finished. best fitness: 473.7614
run 8 finished. best fitness: 472.5849
run 9 finished. best fitness: 468.1095
run 10 finished. best fitness: 472.5849

==================== results ====================
absolute best route length: 468.109452
number of successful runs (<= 480): 9 of 10
```

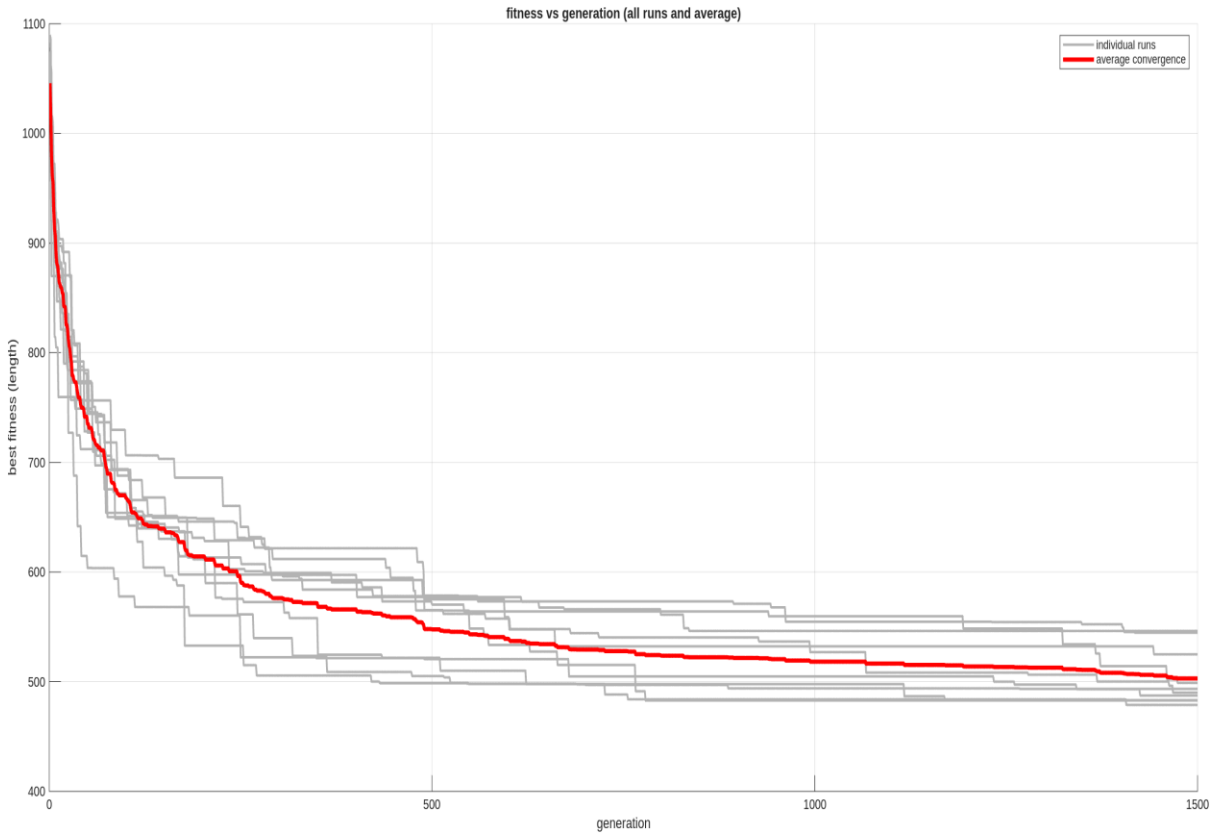
Without swapgen

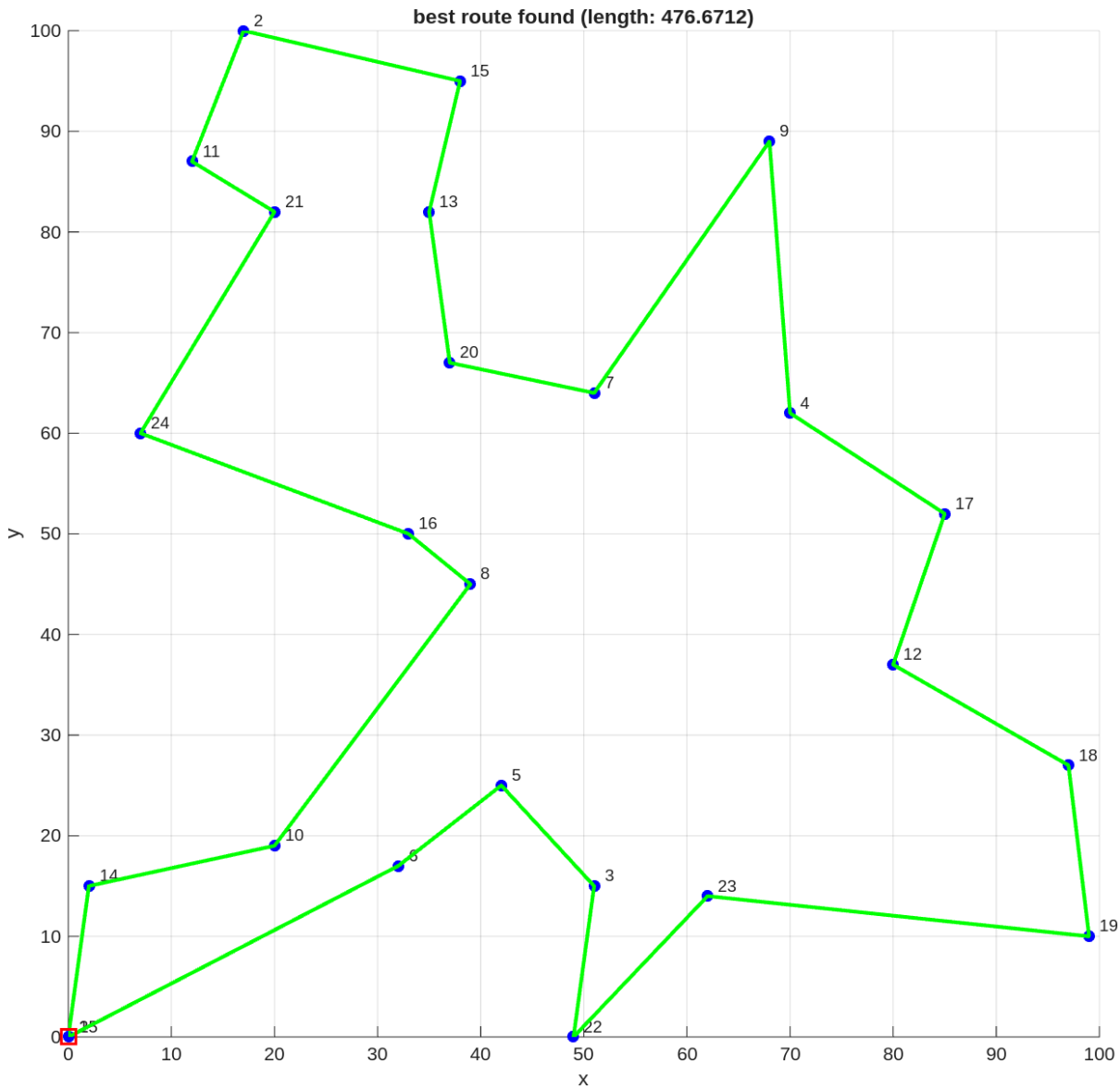

fitness vs generation (all runs and average)
individual runs
average convergence
best fitness (length)
generation
1100
1000
900
800
700
600
500
400
0
500
1000
1500


best route found (length: 468.1095)
x
y
2
15
11
21
13
9
20
7
4
24
17
16
8
12
5
18
10
6
14
3
23
19
25
22

```
popSize = 100;
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 1500;
numRuns = 10;

starting calculations (10 runs)...
run 1 finished. best fitness: 476.6712
run 2 finished. best fitness: 472.5849
run 3 finished. best fitness: 483.3271
run 4 finished. best fitness: 472.5849
run 5 finished. best fitness: 483.1013
run 6 finished. best fitness: 468.1095
run 7 finished. best fitness: 472.5849
run 8 finished. best fitness: 473.7614
run 9 finished. best fitness: 468.1095
run 10 finished. best fitness: 472.5849

==================== results ====================
absolute best route length: 468.109452
number of successful runs (<= 480): 8 of 10
```

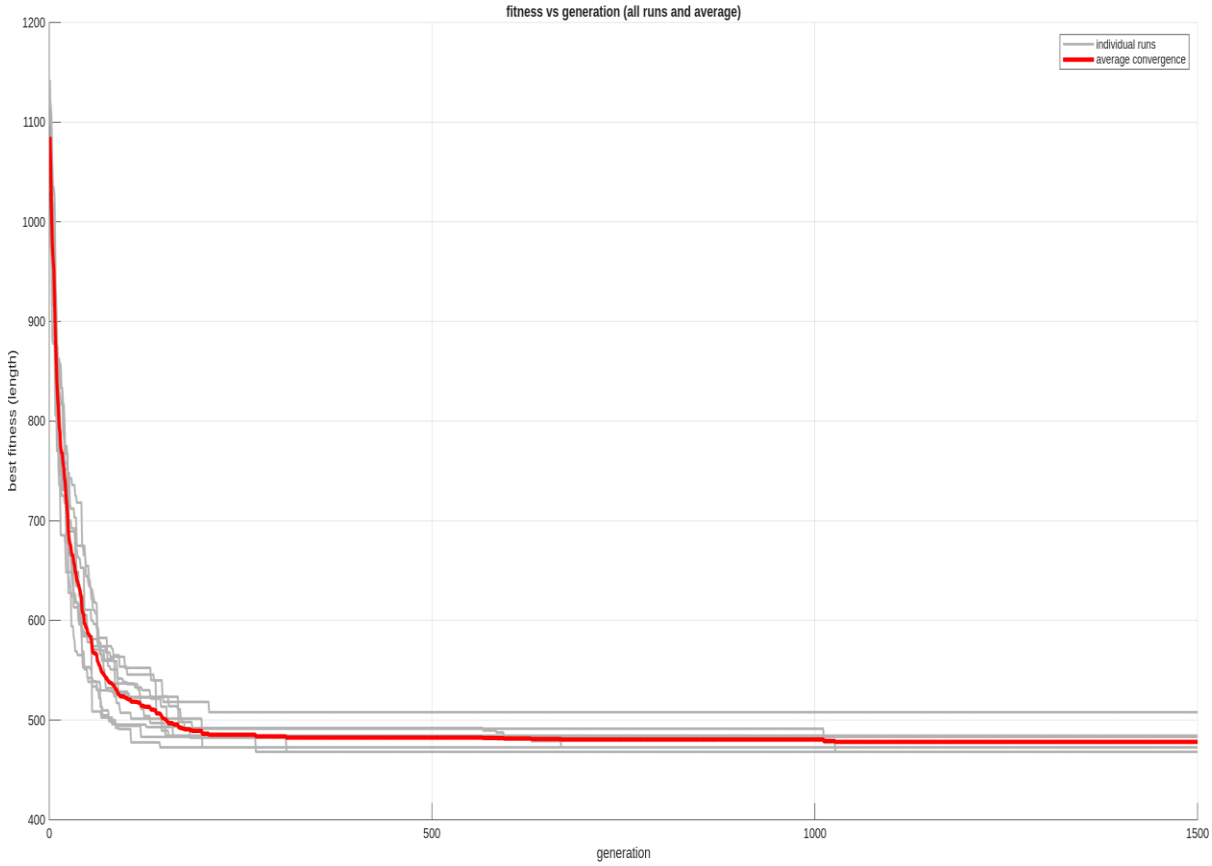
MutRate = 0.9

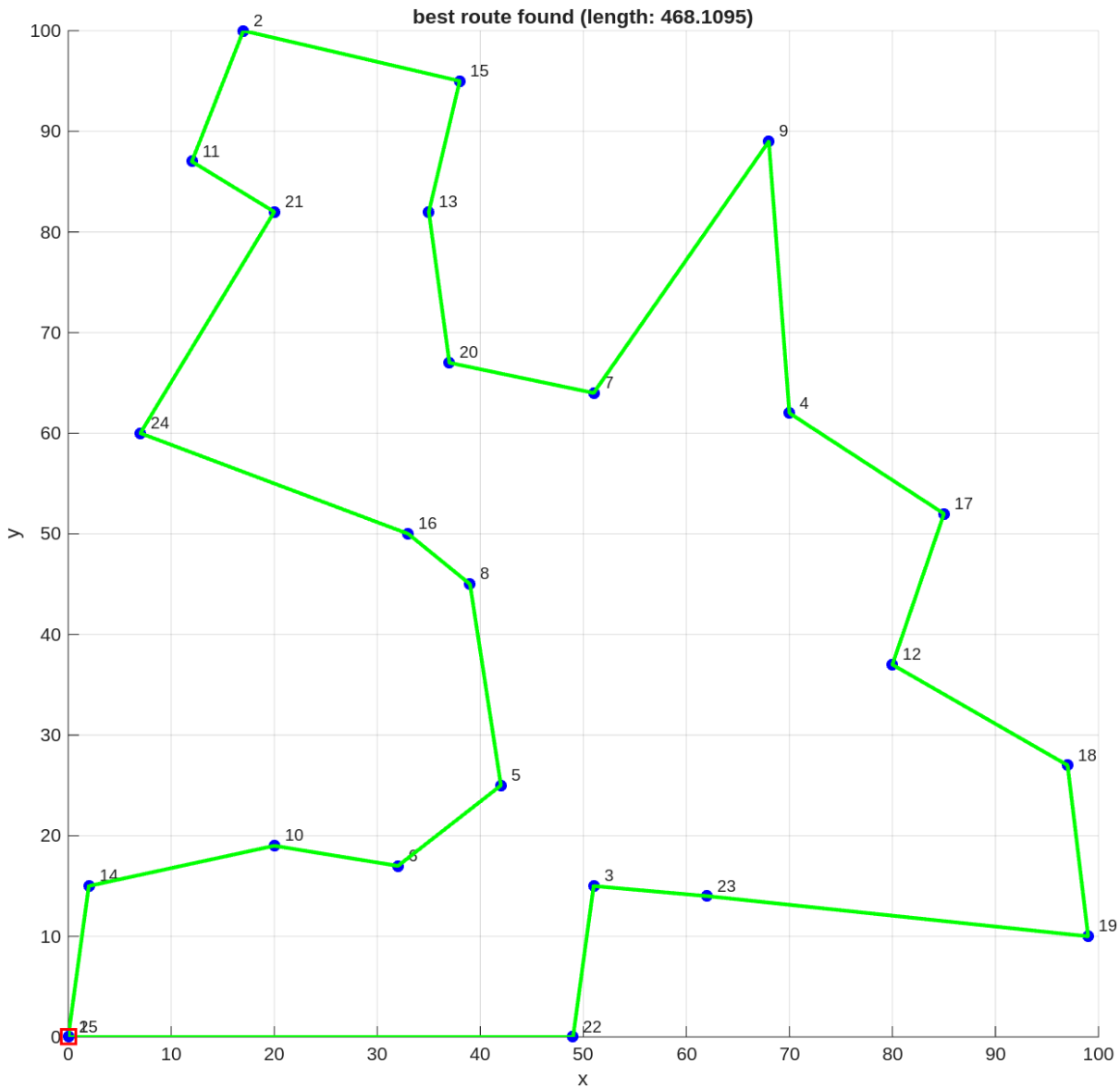





```
popSize = 100;
elitismCount = 5;
mutationRate = 0.9;
numGenerations = 2500;
numRuns = 10;
starting calculations (10 runs)...
run 1 finished. best fitness: 505.1680
run 2 finished. best fitness: 477.2182
run 3 finished. best fitness: 476.6712
run 4 finished. best fitness: 520.0965
run 5 finished. best fitness: 553.5806
run 6 finished. best fitness: 492.4358
run 7 finished. best fitness: 526.1795
run 8 finished. best fitness: 487.3989
run 9 finished. best fitness: 524.6329
run 10 finished. best fitness: 491.7172

==================== results ====================
absolute best route length: 476.671241
number of successful runs (<= 480): 2 of 10
```

PopSize = 50


fitness vs generation (all runs and average)
individual runs
average convergence
best fitness (length)
generation
1200
1100
1000
900
800
700
600
500
400
0
500
1000
1500




```
popSize = 50;
elitismCount = 5;
mutationRate = 0.1;
numGenerations = 1500;
numRuns = 10;

starting calculations (10 runs)...
run 1 finished. best fitness: 468.1095
run 2 finished. best fitness: 472.5849
run 3 finished. best fitness: 468.1095
run 4 finished. best fitness: 472.5849
run 5 finished. best fitness: 468.1095
run 6 finished. best fitness: 484.3319
run 7 finished. best fitness: 483.1013
run 8 finished. best fitness: 483.1013
run 9 finished. best fitness: 472.5849
run 10 finished. best fitness: 507.7905

==================== results ====================
```

```
absolute best route length: 468.109452
number of successful runs (<= 480): 6 of 10
```